

Edge attributes and performance measures

Supply Chain Edge Attributes		
Sr	Attribute/Feature	Description
1	Link distance	Distance between two nodes (typically by road)
2	Transportation Type	Mode of transport (e.g., Road, Air)
3	Fleet Composition	Type and number of vehicles assigned to the link
4	Maximum Load Capacity	Maximum quantity that can be shipped
5	Minimum Shipment Quantity	Minimum quantity required to start a shipment
6	Lead Time	Time to deliver the shipment from one node to another
7	Product(s)	List of products (SKUs) that are being transported
8	Transportation cost	Cost to transport goods from one node to another in the SC
9	Carbon Cost	Cost associated with carbon emissions from transport on this link
10	Payment Terms	Credit period or payment deadline agreed between the two nodes for orders on the edge, as modeled in the resilience cluster for financial disruptions
11	Probability of Link Failure	Probability of this link being unavailable due to natural disasters, road maintenance, or road blocks (e.g., rallies)
12	Disruption Duration	Time for which the link remains unavailable after a disruption
13	Recovery Duration	Time taken by the link to resume service after a disruption
14	Disruption Cost	Cost incurred per disruption on this link, including the cost of recovery
Supply Chain Edge Performance Measures		
Sr	Performance Measure	Description
1	Material Flow	Quantity of product units transported over the link in a given time period
2	CO2 cost	Cost associated with carbon emissions from transportation
3	CO2 emissions	Quantity of carbon emissions from transportation on the edge
4	Link utilization	Utilization rate of the transportation link (e.g., % usage vs. capacity)
5	Total Shipments	Number of shipments sent or received over the link
6	Edge resilience	Resilience depends on the reliability of the edge and connected nodes
7	Edge criticality	Refers to the importance of the edge